GAMIFIED VR EDUCATIONAL PLATFORM FOR MIDDLE SCHOOL CURRICULUM IN SRI LANKA

25-26J-423

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specializing in Interactive Media

Department of Information Technology

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DECLARATION OF THE CANDIDATE & SUPERVISOR

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

....

Signature of supervisor Date: 2025/08/31

Mr. Didula Chamara Thanaweera Arachchi

....

Signature of co-supervisor Date: 2025/08/31

Mr. Nushkan Nismi

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ABSTRACT

This proposal outlines the design of a Virtual Reality (VR)-based gamified educational module focused on engineering, set in an ancient Sri Lankan village near Sigiriya. The module aims to teach secondary level STEM concepts through immersive historical simulations. The background emphasizes the importance of immersive learning for STEM education and identifies a gap in contextualized engineering learning tools in Sri Lanka. The proposed solution addresses low student engagement in engineering subjects by leveraging VR's ability to enhance motivation, conceptual understanding, and knowledge retention. In the simulated historical village environment, students will perform interactive tasks (e.g., constructing structures, managing water systems, using simple machines) that map to the local engineering curriculum. The objectives include increasing student engagement, improving visualization of engineering concepts, and aligning content with Sri Lankan Grade 10–11 engineering syllabi. The methodology involves an iterative development process: curriculum analysis, 3D modeling of the village and artifacts, integration of gamified tasks, and user testing with feedback loops. Key technologies include Unity for VR development, Blender for 3D modeling, and standalone VR headsets for deployment. The project requirements detail functional features (such as virtual interactions, quizzes, and feedback) and nonfunctional criteria (performance, usability, cultural relevance, and safety). A preliminary budget is presented, covering hardware (VR headsets, development workstation) and software resources, along with justification for each item. By situating engineering education in a familiar historical context, this project expects to foster meaningful, experiential learning and improve students' motivation and understanding of engineering principles. The report concludes with a list of references from current research in VR education, gamified learning, Sri Lankan curriculum design, and engineering pedagogy to support the proposed approach.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

Abbreviation	Description
STEM	Science, Technology, Engineering and Mathematics
VR	Virtual Reality
AR	Augmented reality

1. INTRODUCTION

In Sri Lanka, Engineering and STEM subjects are introduced in schools from middle to upper grades, focusing on concepts, problem-solving, and technical skills. However, many schools still rely heavily on textbooks and exam-oriented teaching, leaving students with limited opportunities to apply knowledge in practical or real-life situations. This challenge is especially prominent in rural and under-resourced schools, where students often memorize formulas and definitions but struggle to visualize or experiment with engineering concepts. Feedback from our informal survey at Halwathura K.V. and Sri Dharmaloka K.V. revealed that students often find lessons abstract, and struggle to connect theoretical concepts with practical applications.

Virtual Reality (VR) offers an innovative way to address these challenges by creating immersive and interactive learning environments. Instead of being passive learners, students can step into virtual worlds where they must apply engineering principles to solve problems, design structures, and interact with virtual tools and components. Previous studies show that VR, especially when combined with gamification elements like points, levels, and rewards, increases engagement, conceptual understanding, and knowledge retention compared to traditional teaching methods. This suggests that a gamified VR module could make STEM learning more hands-on, meaningful, and enjoyable, particularly for students who currently find the subjects intimidating.

This project introduces such a module as part of a larger VR platform integrating STEM subjects, including Engineering, Science, and Mathematics. Each game level is set in a historical or culturally familiar Sri Lankan context, blending heritage with practical STEM tasks. For Engineering, students might practice principles of construction, materials selection, or mechanical problem-solving through story-driven missions that simulate real-life projects.

In short, this VR STEM learning module aims to transform traditional lessons into interactive, experiential learning experiences. By situating STEM education in culturally relevant scenarios and combining it with gamified motivation, the project seeks to bridge the gap between theoretical knowledge and practical application, helping students build confidence in applying STEM concepts effectively.

1.1 Background and Literature Survey

Immersive learning environments have gained prominence as effective tools for science and engineering education. Virtual Reality (VR) offers interactive, firstperson learning experiences that traditional classrooms often lack. Research indicates that VR-based learning significantly enhances student engagement, motivation, and conceptual understanding in STEM fields. In engineering education in particular, where many concepts are abstract and process-oriented, VR enables students to visualize structures, systems, and mechanisms in three-dimensional space, leading to stronger knowledge retention than passive learning methods. A recent systematic review on VR use in secondary education highlighted a strong global focus on STEM subjects and reported that VR interventions improved both student motivation and understanding of complex concepts. By simulating realistic scenarios, VR allows students to safely practice engineering skills and apply theoretical knowledge without real-world risks, supporting active, problem-based learning approaches beyond textbook-based instruction.

In parallel, gamified learning strategies have proven effective in increasing student motivation and improving learning outcomes. Gamification involves integrating game elements such as points, challenges, feedback, and rewards into educational activities. Studies show that gamified learning environments encourage sustained engagement and increased time-on-task by transforming lessons into interactive challenges. When combined with VR, gamification creates "serious games" that merge immersion with enjoyment. A 2023 systematic review found that gamified approaches in secondary and higher education consistently had a positive impact on student motivation, particularly in STEM subjects. This project applies these findings by designing the engineering module as a gamified VR experience, with the expectation of improving student engagement, persistence, and conceptual understanding of engineering principles.

Sri Lanka's education system has recently begun exploring immersive technologies as part of its modernization of STEM education. The Ministry of Education has recognized the need to strengthen practical and applied learning in engineering and technical subjects and has piloted VR-based learning initiatives in schools. Under the *13 Years Guaranteed Education* program, VR headsets were deployed to schools with content for technical and vocational subjects. By 2025, over 460 schools received VR equipment with curriculum-aligned content in areas such as agriculture, design, and engineering. While this initiative demonstrates national interest in immersive learning, the existing content largely consists of passive 360° videos and basic demonstrations. There remains a clear need for fully interactive, curriculumaligned VR modules that actively engage students in engineering problem-solving and decision-making.

1.2 Research Gap

Despite global advancements in VR education, a clear gap exists in applying these technologies to engineering learning within Sri Lanka's local context. Traditional teaching of school-level engineering or technology subjects in Sri Lanka largely relies on didactic methods and static content. Students learn principles of mechanics, drawing, and circuits in abstract, disengaged ways, and there are few opportunities to visualize or practice these concepts interactively. While some schools have started using computer simulations, they often depict foreign scenarios or are not aligned with the Sri Lankan curriculum. There is currently no VR-based gamified module that is explicitly designed around Sri Lankan educational standards and cultural settings. In particular, no solution integrates engineering education with Sri Lanka's historical heritage to create an immersive learning experience. This absence is noteworthy because Sri Lanka's engineering achievements (ancient irrigation, architecture, etc.) are ideal context for illustrating basic engineering principles in a relatable way. Another aspect of the gap is the lack of curriculum alignment and local language support in existing VR educational content. The VR experiences introduced in schools via the ClassVR initiative, for example, included generic content and 360° media that were curated to fit broad subject areas, but not purpose built for the national syllabus. Teachers still have to make connections between these experiences and the specific topics they need to cover. Our module seeks to fill this gap by explicitly mapping VR tasks to the local Grade 10–11 engineering technology curriculum (e.g., topics from Design & Mechanical Technology, Construction Technology, etc.). By doing so, we ensure that the virtual activities directly reinforce syllabus content and learning outcomes. Moreover, by providing instructions and narrative in the students' language (Sinhala/Tamil, with English as needed), we make the experience accessible and comfortable, something lacking in many imported educational apps

2. RESEARCH PROBLEM

In the Sri Lankan secondary education system, Engineering Technology is introduced with the intention of developing students' technical knowledge, problemsolving ability, and practical understanding of engineering principles. However, in practice, the teaching and learning process remains largely theory-driven and examination-oriented. Engineering concepts such as structural stability, material properties, mechanical systems, and construction processes are often delivered through textbooks, diagrams, and verbal explanations, with limited opportunities for hands-on experimentation or real-world application. As a result, many students struggle to visualize abstract concepts and to connect theoretical knowledge with practical engineering situations.

This challenge is further intensified by constraints faced by schools, particularly in rural and under-resourced areas, where access to laboratories, tools, and physical models is limited. Without practical exposure, students tend to memorize definitions and procedures rather than develop a deep conceptual understanding of engineering principles. This leads to low engagement, reduced motivation, and difficulty in applying engineering knowledge to problem-solving tasks. Consequently, students may perceive Engineering Technology as difficult or uninteresting, despite its relevance to real-life applications and national development.

Although digital learning tools and simulations have been introduced in some educational settings, most existing resources are either passive in nature, based on foreign contexts, or not directly aligned with the Sri Lankan Engineering Technology curriculum. Additionally, these tools rarely incorporate culturally relevant examples that could help students relate engineering concepts to their own historical and social environment. There is a noticeable absence of immersive, interactive learning solutions that allow students to actively experiment, make mistakes, receive immediate feedback, and learn through experience in a safe and controlled environment.

Therefore, the core research problem addressed in this study is how a VR-based gamified engineering learning environment can be designed and implemented to effectively support the teaching of Engineering Technology concepts for Sri Lankan students. Specifically, the study seeks to explore how immersive VR experiences, combined with gamification and historically contextualized engineering scenarios, can enhance student engagement, improve conceptual understanding, and ensure alignment with the Grade 10–11 Sri Lankan Engineering Technology curriculum, while also respecting local cultural and educational contexts.

3. OBJECTIVES

3.1 Main Objective

To design and develop a Virtual Reality (VR)-based gamified engineering education module aligned with the Sri Lankan Grade 10–11 Engineering Technology curriculum, with the aim of improving student engagement and conceptual understanding of engineering principles.

3.2 Specific Objectives

- To align VR-based learning activities with the learning outcomes of the Grade 10–11 Engineering Technology syllabus in SriLanka.
- To integrate historical Sri Lankan engineering applications as contextual learning scenarios within the VR environment.
- To enhance students visualization and understanding of abstract engineering concepts through immersive and interactive VR experiences.
- To incorporate gamification elements such as feedback, challenges, and progression to increase student motivation and engagement.
- To evaluate the educational effectiveness of the VR module in terms of student engagement and conceptual understanding.

4. METHODOLOGY

The methodology of this project was designed to ensure that the development of the Engineering VR module followed a structured and systematic process. It consisted of three key phases: curriculum and user needs analysis, module development, and testing and evaluation.

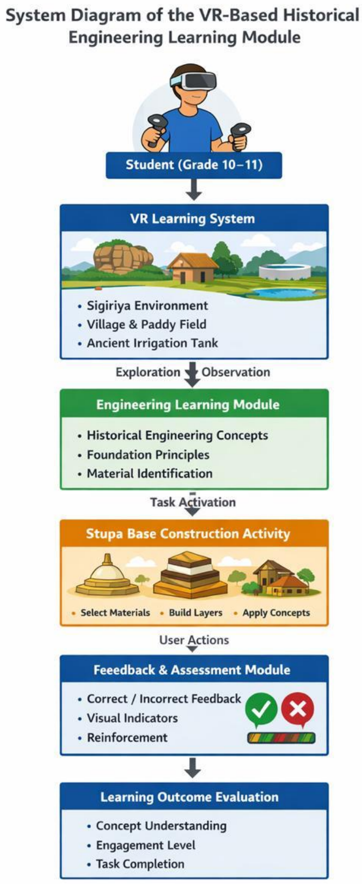

Figure 4.1 - High-Level System Diagram

Needs Analysis & Curriculum Mapping

The first stage involved analyzing the Grade 7-9 English curriculum in detail to identify the key language skills students are expected to acquire, including vocabulary, grammar structures, reading comprehension, and sentence construction. To complement this review, surveys and interviews were conducted with English teachers and students from Halwathura K.V. and Sri Dharmaloka K.V. schools. These interactions revealed common student difficulties, such as weak vocabulary retention, limited understanding of grammar rules, and lack of exposure to spoken English.

The insights gathered informed the selection of target syllabus content for the VR module. For instance, vocabulary themes were chosen directly from the Grade 7-9 textbooks, while grammar activities were modeled on the syllabus-prescribed patterns and structures. The analysis ensured that the VR module would not deviate from classroom requirements but would instead reinforce them in a more interactive way.

Measurable Outcome: A structured list of syllabus-aligned vocabulary themes, grammar structures, and comprehension tasks to be embedded into the VR module.

Module Development in Unity 3D

In the second phase, the English module was developed using Unity 3D for the Meta Quest 3 VR headset. The development process included the following components:

- **3D Environments:** Historical scenes created by the History team (e.g., ancient marketplaces, temples, colonial-era settings) were integrated as the backdrop for English tasks.
- **Interactive Tasks:** Activities such as multiple-choice dialogues, vocabulary labeling of items in scenes, grammar-based sentence reordering puzzles, and short comprehension passages were programmed to encourage active participation.
- **Bilingual Support:** Sinhala-English narration and subtitles were added to make instructions accessible while still prioritizing English practice.
- **Gamification Features:** A points system, badges, and progress indicators were implemented to motivate learners. Completing tasks successfully allowed students to unlock new levels and scenes.
- **Data Logging:** A built-in tracking system recorded student progress, task completion rates, and performance accuracy for later evaluation.

The development process followed an iterative cycle, where prototypes were tested internally, refined based on feedback, and gradually improved.

Testing & Evaluation

The final phase of the methodology focuses on evaluating the VR module's effectiveness. Pilot testing is planned with Grade 7–9 students from partner schools. The evaluation process will involve both qualitative and quantitative methods:

- **Pre- and Post-Tests:** To measure vocabulary retention, grammar accuracy, and comprehension improvements.

- **Observation and Feedback:** Classroom sessions will be observed to assess student engagement and interaction with the module. Teachers and students will provide feedback on usability and relevance.
- **Motivation and Confidence Surveys:** Students will complete questionnaires to capture changes in their attitude toward English learning and their confidence in using the language.
- **Feasibility Assessment:** Practical issues such as hardware availability, student adaptability to VR, and teacher readiness will be documented to evaluate scalability.

Outcome of Phase 3: Evidence of learning improvements, increased engagement, and practical recommendations for scaling the VR solution in Sri Lankan schools.

5. PROJECT REQUIREMENTS

5.1 Functional Requirements

- **Curriculum-Aligned Content Integration**

  The VR module must incorporate Engineering learning activities directly aligned with the Grade 10-11 national syllabus in Sri Lanka. The module should also allow easy integration of additional content for future curriculum updates.

- **Immersive 3D Historical Environments**

  Students will interact in 3D environments designed around Sri Lanka's historical settings (e.g., ancient marketplace, temple, colonial classroom). These environments will be interactive, allowing learners to complete tasks by selecting objects, engaging in dialogues, or solving contextual puzzles.

- **Interactive Engineering Tasks**

  The platform must support interactive activities.

- **Gamification and Progress Tracking**

  The system must implement points, badges, and progress indicators for each level. Rewards such as unlocking the next scene or earning badges will keep learners motivated. Progress will be tracked per user, and teachers will be able to review student achievement summaries.

5.2 Non-Functional Requirements

- **Reliability** - The VR module must run consistently on Meta Quest 3 headsets without frequent crashes, ensuring uninterrupted learning sessions.

- **Performance** - The system should load immersive scenes quickly (within 5 seconds) and allow smooth interaction (60+ FPS in VR). Student tasks and feedback should appear instantly to maintain engagement.
- **Usability** - The system should be intuitive for middle school learners, with simple navigation menus and minimal training required for teachers.

5.3 System Requirements

Hardware Requirements

- **Development**: A workstation with a multi-core CPU, 16 GB RAM minimum, and a GPU (NVIDIA RTX 3060 or higher) for Unity 3D scene rendering.
- **Deployment**: Meta Quest 3 headset with minimum 128 GB storage. Lightweight classroom PCs or mobile devices may be used for teacher dashboards.

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Software Requirements

- **Development Tools**: Unity 3D, Blender (for 3D modeling), Visual Studio.
- **VR SDKs**: Meta Quest SDK, Oculus Integration for Unity.
- **Supporting Libraries**:
  - Localization tools for text and audio.
  - Database system (SQLite/PostgreSQL) for user progress tracking.
  - Analytics libraries for usage monitoring.

User Interface Requirements

- Student UI (in VR):
  - Easy navigation with VR controllers.
  - Clear visual cues for interactive tasks (highlighted objects, arrows).
  - Gamified progress bar and task completion feedback..

- Teacher/Admin UI (outside VR):
  - Web or desktop dashboard to upload/update tasks.
  - Ability to track student performance (scores, progress, areas of difficulty).
  - Exportable reports (CSV/PDF) for academic records.

Performance Requirements

The VR Engineering learning module is designed to deliver a smooth and immersive experience for middle school learners. To achieve this, the application must maintain a stable frame rate of 60-72 FPS, which is essential for preventing motion sickness and ensuring comfort during extended use. Each new scene within the historical environments should load within 5-7 seconds to maintain engagement and avoid unnecessary delays. Activities within the module are structured so that learners can complete them within 2-3 minutes, striking a balance between challenge and flow. Additionally, the analytics system must be capable of processing learner progress data in real time, storing it securely without noticeable delays, so that both students and teachers receive timely feedback.

Compatibility Requirements

The VR module will be fully optimized to run natively on Meta Quest 3 headsets, ensuring compatibility with modern standalone VR hardware. For teachers and administrators, the supporting dashboards will be designed to be cross-platform, accessible on both Windows and Linux operating systems, and compatible with widely used web browsers. The system will also support data input and output in standard formats such as CSV, JSON, and Excel, making it easier to integrate with existing tools. Furthermore, the system architecture will be modular and scalable, allowing for future interoperability with school Learning Management Systems (LMS) and the integration of new features without major redevelopment.

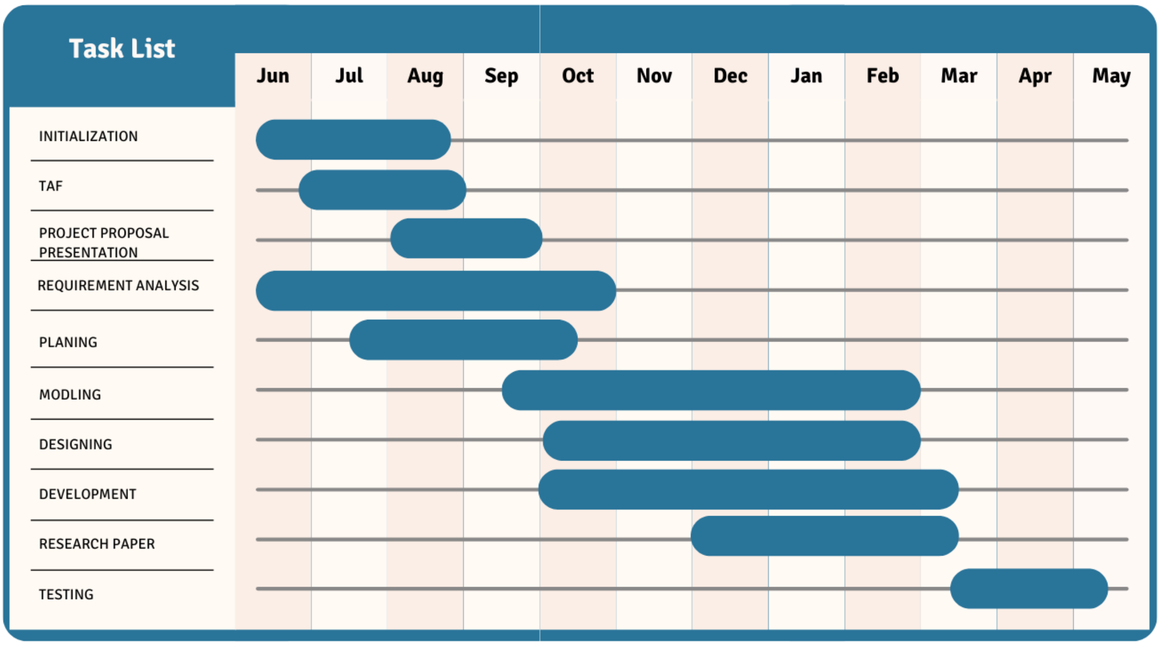

Figure 4.2 - Work brake down structure

6. BUDGET AND BUDGET JUSTIFICATION

The below table depicts the budget required for the entire proposed system with other related payments.

Item	Estimated Cost (LKR)	Justification
Meta Quest 3 Headset	0 (already available)	Main hardware for developing and testing the VR platform. Quest 3 is portable and doesn't need a PC in the classroom, making it ideal for schools. One unit is enough for development and demo purposes.
PC for Development	0 (already available)	Using an existing PC from the university lab or personal use. Meets Unity VR development needs, so no extra cost.
Unity Pro/Plus License	0	Unity free version has all needed features for students. No paid license required.
Unity Asset Store Purchases	10,000	High quality 3D models (temples, colonial buildings, vegetation) to make worlds realistic and save time. Will use free assets first; purchase only if needed.
Voice Recording Setup	0	Use university's audio lab and volunteer voices for narration in Sinhala and English. No extra cost expected.
Testing & Deployment Costs	5000	Transport to partner schools for pilot testing, printing quick start guides for teachers/students, and other small expenses.
Contingency	10,000	Backup for unexpected costs, such as replacing a controller strap or purchasing an essential software plugin.

Table 6.1 - Budget & Budget Justification

Total Estimated Budget: LKR 25,000

7. COMMERCIALIZATION PLAN

To address the limitations of traditional classroom-based Engineering Technology education and the growing need for practical, application-oriented learning, the proposed system a Gamified VR Engineering Education Module for Sri Lankan secondary school students (Grades 10–11) has been developed. By combining immersive Virtual Reality environments, curriculum-aligned engineering tasks, and historically contextualized problem-solving scenarios, the platform offers an engaging, scalable, and cost-effective solution for teaching engineering concepts. The commercialization strategy aims to make this technology accessible to public and private schools, technical colleges, and educational institutions, thereby supporting improved student engagement and understanding of engineering principles.

The proposed VR module is designed to complement existing classroom teaching rather than replace it, making it suitable for gradual adoption within the Sri Lankan education system. The solution can be deployed using standalone VR headsets, reducing infrastructure requirements and enabling use in both urban and rural schools. Through commercialization, the project seeks to bridge the gap between theoretical engineering instruction and practical skill development by providing students with an interactive learning experience aligned with national curriculum requirements.

An official product website and digital outreach platform will be launched to present the features, educational benefits, and practical use cases of the VR Engineering Module. Through this platform, schools and educators will be able to explore demonstrations, access sample learning scenarios, request pilot implementations, and initiate purchase or partnership discussions. Establishing a strong digital presence will help build credibility, raise awareness among stakeholders, and accelerate adoption within the education and training sectors.

Target Audience

The primary target audience for the VR Engineering Education Module includes:

- **Schools (Public and Private):** Secondary schools offering Engineering Technology subjects and seeking innovative tools to enhance practical learning and student engagement.
- **Engineering Technology Teachers and Curriculum Planners:** Educators looking for interactive teaching aids to support syllabus topics such as materials, structures, simple machines, and construction processes.
- **Educational Administrators and Policymakers:** Decision-makers at school, zonal, and ministry levels who aim to improve STEM education quality, practical skill development, and student interest in engineering-related pathways.

- **Technical Colleges and Vocational Training Institutes:** Institutions that require safe, cost-effective simulation tools to introduce foundational engineering concepts before hands-on training.
- **Educational NGOs and STEM Promotion Programs:** Organizations working to increase STEM awareness and access, particularly in rural or underresourced regions.

While students are the direct beneficiaries of the system, teachers and institutions gain value through enhanced engagement, improved conceptual understanding, reduced dependence on physical lab resources, and data-driven insights into learner performance.

Marketing Strategy

The marketing strategy will position the VR Engineering Education component as a next-generation STEM learning tool that combines technology, curriculum relevance, and cultural context. The approach will primarily follow a B2B and B2B2C model, focusing on schools and educational institutions, supported by awareness initiatives targeting educators and education authorities. Key strategies include:

- **Pilot Deployments:** Implementing the VR module in selected urban and rural schools to gather feedback, validate educational impact, and develop case studies demonstrating real classroom benefits.
- **Educational Conferences and Exhibitions:** Showcasing the module at national and regional education, STEM, and EdTech events to highlight its innovative use of VR for engineering education.
- **Institutional Partnerships:** Collaborating with the Ministry of Education, curriculum development bodies, teacher training institutes, and EdTech distributors to support wider adoption and curriculum integration.
- **Research and Academic Publications:** Publishing findings from pilot studies and evaluations in academic and professional journals to establish educational effectiveness and strengthen credibility.
- **Digital Outreach and Demonstrations:** Creating promotional videos, interactive demos, and testimonials to be shared through social media platforms, educational websites, and the official product site to reach educators, administrators, and parents.

The marketing narrative will emphasize how the VR Engineering Module addresses key challenges in engineering education such as abstract concept visualization, limited practical exposure, and low student engagement by offering immersive, interactive, and gamified learning experiences aligned with the Sri Lankan Engineering Technology curriculum. By highlighting both academic outcomes (improved conceptual understanding and application skills) and educational benefits (increased motivation and interest in engineering) the platform aims to differentiate itself from traditional teaching methods and generic VR applications.

8. CONCLUSION

The VR-based gamified engineering education module has been designed to directly address the persistent challenges faced by Sri Lankan secondary school students in learning Engineering Technology concepts. By combining immersive Virtual Reality environments with interactive, gamified engineering tasks, the module provides an engaging and experiential learning platform where students can actively explore, experiment, and apply engineering principles in meaningful contexts. Unlike traditional theory-driven and rote-based teaching methods, this approach encourages learning through doing, allowing students to visualize abstract concepts, make decisions, and learn from mistakes in a safe virtual environment.

Aligned with the Grade 10–11 Sri Lankan Engineering Technology curriculum, the module reinforces key syllabus topics such as materials, structures, construction processes, and simple mechanical systems through historically contextualized problem-solving activities. The inclusion of real-time feedback and performance tracking supports continuous learning and helps students develop a deeper conceptual understanding of engineering principles. By situating learning within a simulated ancient Sri Lankan village, the module also strengthens cultural relevance and demonstrates how engineering concepts have been applied in local historical contexts.

In conclusion, this engineering component demonstrates the potential of Virtual Reality and gamification to transform Engineering Technology education into a more effective, engaging, and learner-centered process. The proposed module lays the foundation for a scalable and adaptable educational solution that can be integrated into schools across Sri Lanka, contributing to improved student motivation, stronger conceptual understanding, and increased interest in engineering-related STEM pathways.

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